

Course Syllabus

Course Title	Logic Circuits	Course Code	ELEC0247-010	Credits	3-3-0
Department	College of IT Engineering Department of Electrical Engineering	Term/Year	2025 1st Semester	Course Categories	Major in Engineering
Instructor	알버트폴로즈	Class Hour	Mon. 1A,1B,2A Fri. 2B,3A,3B	Classroom	공대8호관 201
Pone	01043890577	Email	albertpoulosepalatty@gmail.com	Classroom Language	English
Office& Office Hours	Appointment in advance by email				
Talent Model of department	Creative global energy leader for the future energy industry				
Educational objective of department	• Cultivation of basic engineering knowledge through rich theoretical education and various practical education across IT • Cultivate creative design ability and system analysis ability for actual electrical and electronic systems • Cultivation of ethical responsibility and cooperative spirit required as an engineer • Cultivation of professional research ability and lifelong learning awareness for leading the rapidly changing information-oriented age • Cultivation of international cooperation ability and on-site practical skills				

1. Course Information

Course Outlin			This course introduces digital logic concepts, including number systems, Boolean algebra, and fundamental logic gates. It covers combinational and sequential logic circuits, focusing on design techniques, minimization methods, and applications like adders, multiplexers, flip-flops, and counters. The course also explores memory devices, programmable logic, and real-world applications in digital systems.					
Core Competencies			Innovativeness		Reflection		Character	
			Creativity ☐	Convergence ☐	Critical Thinking ☐	Exploration ☒	Communication ☐	Responsibility ☐
			Creativity: The ability to "solve problems through a creative approach"as an innovative person Convergence: The competence of "problem solving through convergent thinking and insight"as an innovative person Critical Thinking: The competence of "logical and integrated reasoning through critical thinking"as a reflective person Exploration: The competence of "universal literacy for academic exploration" as a reflective person Communication: The competence of "reasonable communication and aesthetic empathy"as a person of character Responsibility: The competence of "establishing a sense of community and values"as a person of character					
Major	Major Competence							
	performance	knowledge						
		Skill						
	critereion	Attitude						

Course Objectives	1. Gain an understanding of the essential concepts and mathematical principles fundamental to working with digital circuits and systems. 2. Develop professional expertise in analyzing and designing digital circuits and systems, building on a foundational knowledge of logic circuits.							
Prerequisites	None				Recommended Subsequent Courses	Microprocessors, Digital Control		
Grading Scale (100%)	출석	중간시험	기말시험	과제	발표	토론	안전교육	
	10	35	35	0	0	0	0	20
Evaluation Method	- Attendance 10% - Midterm Exam 35% - Final Exam 35% - Homework of each chapter 20% * Homework will be assigned after each chapter through the course homepage. * The solution to the assigned Homework for each chapter will also be provided through the homepage after the quiz.							
Textbook and Other Reference	1. Digital Fundamentals by Thomas L. Floyd 2. Digital Systems: Principles and Applications by Ronald J. Tocci, Neal S. Widmer, and Gregory L. Moss 3. Fundamentals of Logic Design by Charles H. Roth, Jr. and Larry L. Kinney							
Notice to Students	-All the notices related to this course will be on the course homepage. -Attendance will be checked for all the classes. Make-up exams are possible only in exceptional circumstances, such as sickness, family matters, etc. If possible, notify me in advance before the exam. -The final grade is not negotiable for any reason. However, objections with valid reasons in the evaluation are allowed.							
Support Available for Disabled Students	Please inform the professor in advance, as assistance will be provided based on the specific needs of each disability.							

2. Course Content and Schedule

no	Unit Goals and Learning Content	Teaching Methods	Assignments and Research Questions	Note
1	Introductory Concepts	Powerpoint presentations	Selected problems	
2	Number Systems, Operations and Codes	Powerpoint presentations	Selected problems	
3	Logic Gates	Powerpoint presentations	Selected problems	
4	Boolean Algebra and Logic Simplification	Powerpoint presentations	Selected problems	
5	Combinational Logic Analysis	Powerpoint presentations	Selected problems	
6	Midterm Exam			
7	Functions of Combinational Logic	Powerpoint presentations	Selected problems	
8	Latches, Flip-Flops and Timers	Powerpoint presentations	Selected problems	
9	Shift Registers	Powerpoint presentations	Selected problems	
10	Counters	Powerpoint presentations	Selected problems	
11	Topics based on the availability of Time: Programmable Logic	Powerpoint presentations	Selected problems	

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no	Unit Goals and Learning Content	Teaching Methods	Assignments and Research Questions	Note
12	Topics based on the availability of Time: Data Storage, Signal Conversion & Processing	Powerpoint presentations	Selected problems	
13	Topics based on the availability of Time: Data Transmission	Powerpoint presentations	Selected problems	
14	Topics based on the availability of Time: Data Processing & Control	Powerpoint presentations	Selected problems	
15	Final Exam			

3. Course Evaluation

No data
